Grade: Check Plus

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February 23, 2026
Statistics & Data Analysis
Problem set: assignment 2

1. Compute the mean and standard deviation of the stock price.

PRICE (x)	PROBABILITY f(x)	
20	.05	= 1
21	.06	= 1.26
22	.07	= 1.54
23	.08	= 1.84
24	.09	= 2.16
25	.12	= 3
26	.15	= 3.90
27	.13	= 3.51
28	.10	= 2.80
29	.10	= 2.90
30	.05	= 1.50
	1.00	

1+1.26+1.54+1.84+2.16+3+3.90+3.51+2.80+2.90+1.50= 25.41 ✓ Q1

The mean stock price is 25.41 and the standard deviation is $2.74

σ 2 =∑(x−μ) ^2 f(x)

PRICE (x)	Deviation	PROBABILITY f(x)
20	(20-25.41)^2	.05 =
21	(21-25.41)^2	.06 =
22	(22-25.41)^2	.07 =
23	(23-25.41)^2	.08 =
24	(24-25.41)^2	.09 =
25	(25-25.41)^2	.12 =
26	(26-25.41)^2	.15 =
27	(27-25.41)^2	.13 =
28	(28-25.41)^2	.10 =
29	(29-25.41)^2	.10 =
30	(30-25.41)^2	.05 =

✓ Q2a

2. a. What is the probability that the stock price will fall within 1 standard deviation of the mean one year from now? It is a 67% chance.

✓ Q2b

b. What is the probability that the stock price will fall within 2 standard deviations of the mean one year from now? There is a 100% chance.

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3. Derive the distribution of the option value, ignoring interest and commissions.

Stock price	optional value
20-25	0
26	1
27	2
28	3
29	4
30	5

- 20-25 stock price = .47

✓ Q3

- f(v)	optional value
.47	0
.15	1
.13	2
.10	3
.10	4
.05	5

✓ Q4

4. What is the probability that the option will have positive value? 53% chance the option will have positive value.

✓ Q5

5. Compute the mean and standard deviation of the value of the option. The mean is $1.36 and the standard deviation is $1.60

- f(v)	optional value	v x f(v)
.47	0	0
.15	1	.15
.13	2	.26
.10	3	.30
.10	4	.40
.05	5	.25

- 0+.15+.26+.30+.40+.25= 1.36

-	(v-1.36)^2	optional value	f(v)
-	1.85	0	.47
-	.13	1	.15
-	.41	2	.13
-	2.69	3	.10
-	6.97	4	.10
-	13.25	5	.05

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- $\sigma/v\ 2 = \sum(v-\mu v)\ ^2\ f(v) = 2.57$